SENTIMENT ANALYSIS ON IPL TWEETS

1. Problem Statement

With the increasing popularity of social media, users frequently express opinions about live events such as cricket matches. Analyzing these opinions can help understand public sentiment. This project focuses on classifying tweets related to the Indian Premier League (IPL) into positive, negative, and neutral sentiments using machine learning techniques.

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2. Objective

The objectives of this project are:

- To collect and analyze IPL-related tweets
- To preprocess textual data for analysis
- To classify tweets into sentiment categories
- To apply multiple machine learning algorithms
- To compare model performance using evaluation metrics

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3. Dataset

Source:
The dataset consists of 100 manually created IPL-related tweets.

Features:

- Text: Tweet content
- Label: Sentiment (positive, negative, neutral)

Size:

- Total tweets: 100
- Training data: 80 tweets
- Testing data: 20 tweets

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4. Methodology

4.1 Data Preprocessing

- Converted all text to lowercase
- Removed special characters, links, and mentions
- Cleaned noisy text for better accuracy

4.2 Feature Extraction

- Used **TF-IDF (Term Frequency–Inverse Document Frequency)**
- Converted text data into numerical vectors
- Removed stopwords to improve relevance

4.3 Model Building

The following machine learning models were implemented:

- Naïve Bayes
- Logistic Regression
- Support Vector Machine (SVM)

4.4 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

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5. Results

Model	Accuracy	Precision	Recall
Naïve Bayes	~80–85%	~0.80	~0.79
Logistic Regression	~85–90%	~0.86	~0.85
SVM	~88–92%	~0.89	~0.88

Insights:

- SVM performed the best among all models
- Logistic Regression provided balanced performance
- Naïve Bayes was faster but slightly less accurate

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6. Discussion

The results show that machine learning models can effectively classify sentiments from IPL tweets. TF-IDF played a crucial role in improving model performance. The dataset being small limits generalization, but results are satisfactory for academic purposes.

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7. How to Run

1. Install dependencies:

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pip install -r requirements.txt
```

2. Execute the program:

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python main.py
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8. Conclusion

This project successfully demonstrates sentiment analysis using machine learning techniques on IPL-related tweets. Among the models used, SVM achieved the highest accuracy, making it the most effective model for this dataset.

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9. Future Scope

- Use real-time Twitter API data
- Apply deep learning models like LSTM or BERT
- Increase dataset size for better accuracy
- Deploy as a web-based application

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10. Student Details

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